

1 a) if (grade >= 90) System.out.print("Great job!"); b) if (number < 20 number > 50) System.out.print("Error"); c) if (y < 100) y += 2;	2) if (num1 > num2) System.out.println("First number is larger"); else if (num2 > num1) System.out.println("Second number is larger"); else System.out.println("Numbers are equal");
3a) even, odd b) switch (num % 2) { case 0: System.out.println("even"); break; default: System.out.println("odd");	4a) int num = (int)(Math.random() * 50) + 1; b) int num = (int)(Math.random() * 81) + 20; c) double num = Math.random() * 11 + 10;
5) The logic error is that ages 18 and 60 don't display any message.	6a) True b) False c) True d) False e) True f) True g) True
7a) y = Math.pow(y, 2); b) y = Math.abs(y); c) y = Math.sqrt(y);	8a) True b) False: a nested if and an if-else are different structures. c) False: a switch expression must evaluate to an appropriate integral type, String, enum, etc., not necessarily a double. d) True e) False: casting is not required to generate a random integer; you can use casting with Math.random(), but other methods exist. f) True g) True h) True i) False: Math.pow() is used for exponentiation. j) False: x = 3 assigns 3 to x; it does not return a value.